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(54) **DISPOSABLE COMPONENTS FOR
REUSABLE PHYSIOLOGICAL SENSOR**

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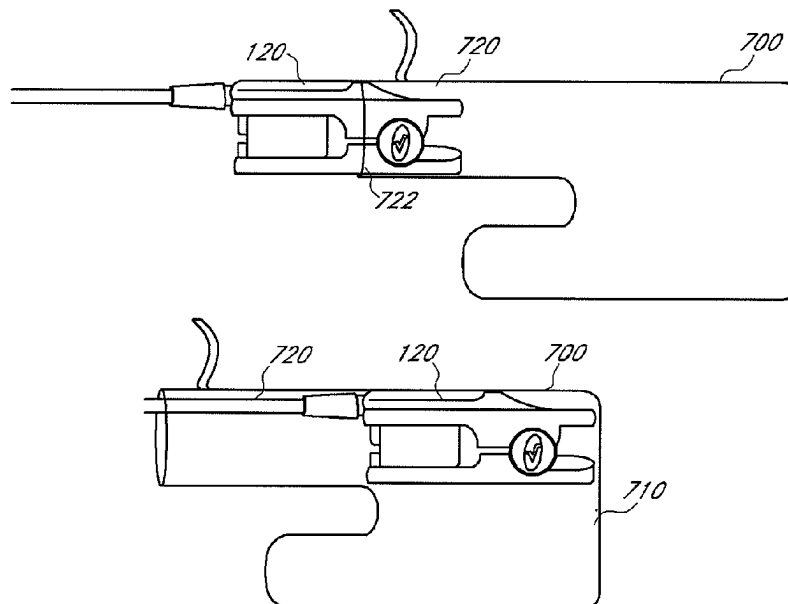
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(57) **ABSTRACT**

A sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor. Certain embodiments of the sensor cartridge protect the sensor from damage, such as damage due to repeated use, reduce the need for sensor sanitization, or both. Further, embodiments of the sensor cartridge are positionable on the user before insertion in the sensor and allow for improved alignment of the treatment site with the sensor. In addition, the sensor cartridge of certain embodiments of the disclosure can be configured to allow a single sensor to comfortably accommodate treatment sites of various sizes such as for both adult and pediatric applications.

9 Claims, 13 Drawing Sheets



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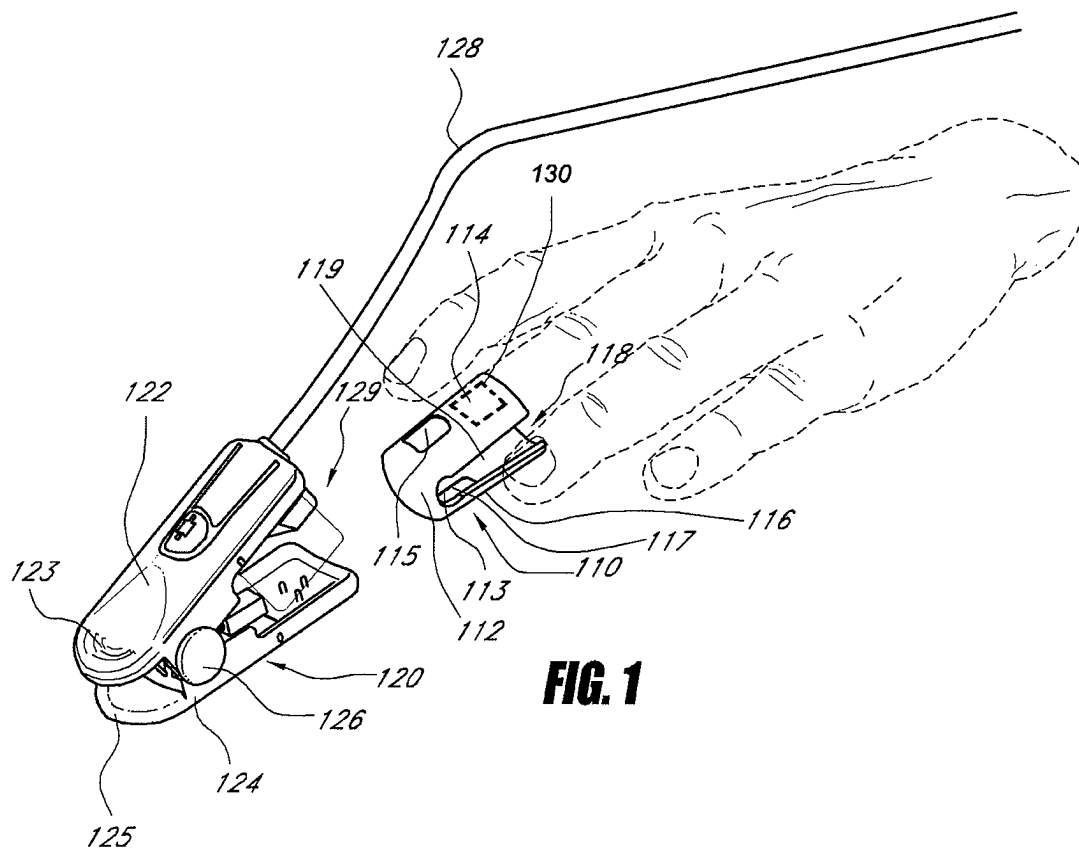
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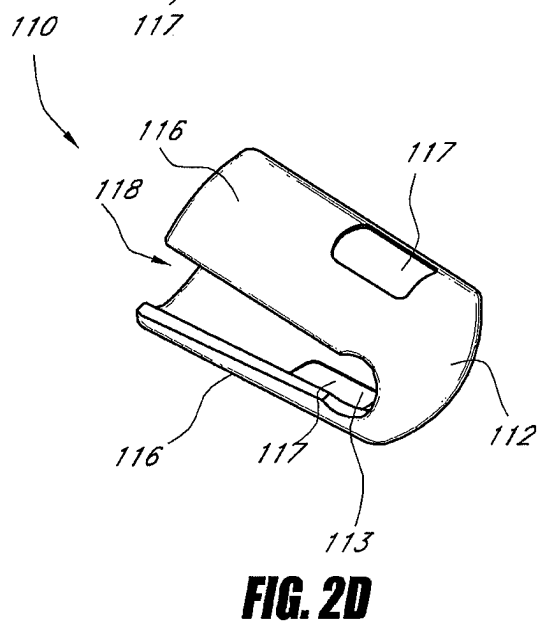
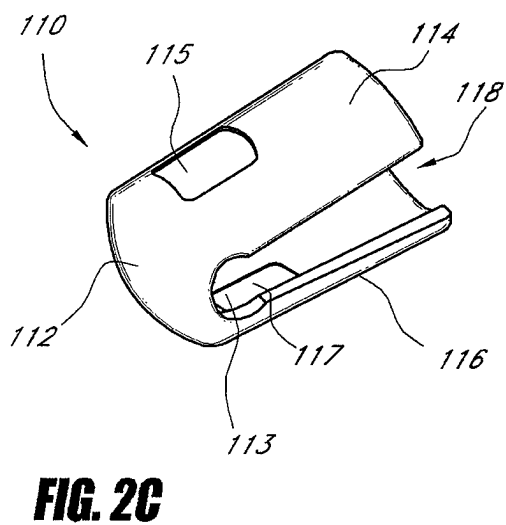
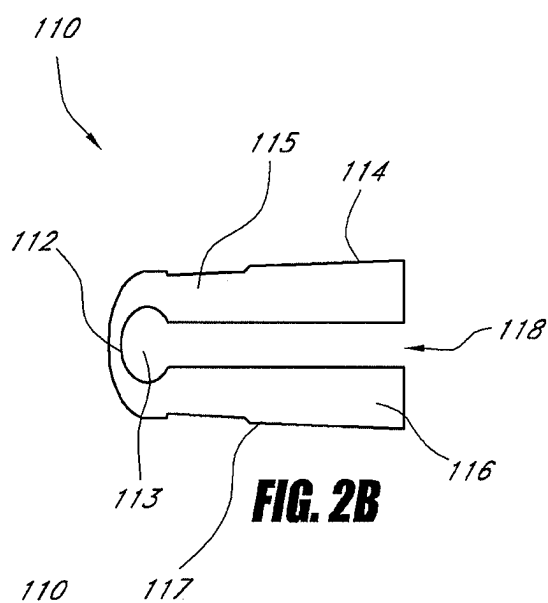
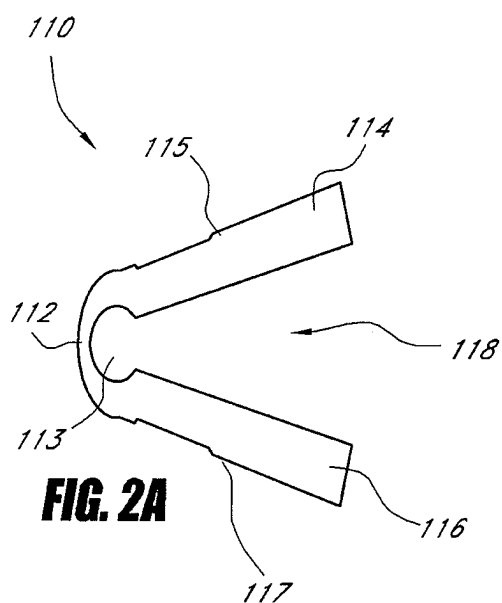
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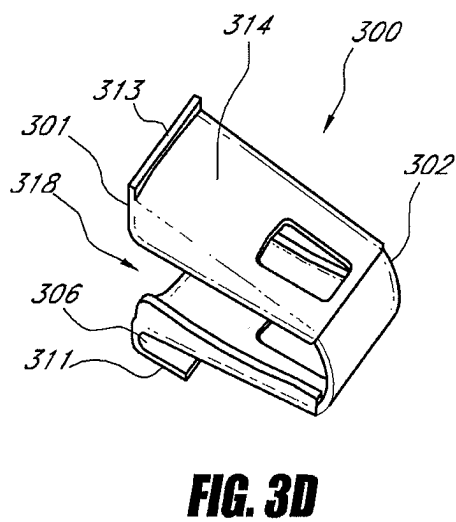
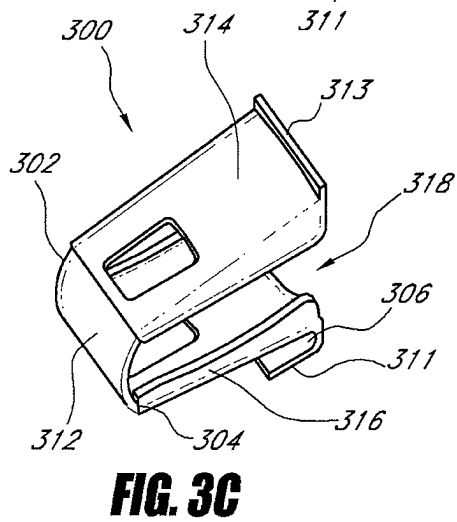
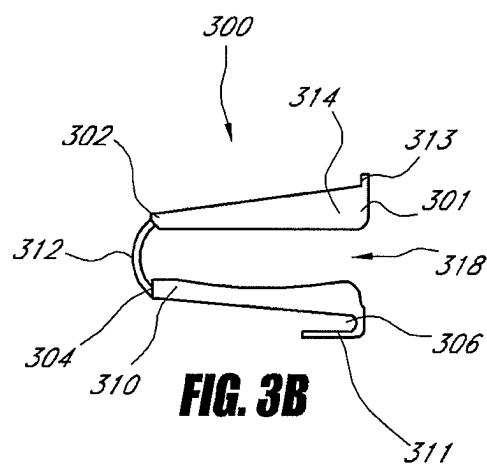
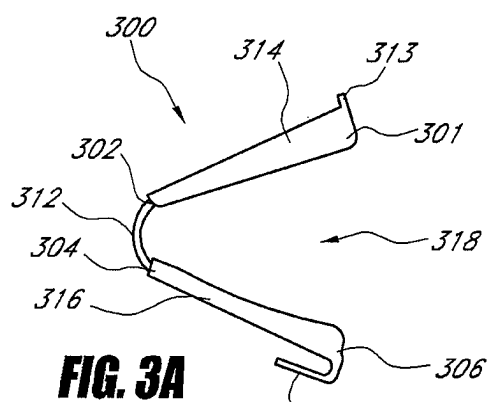
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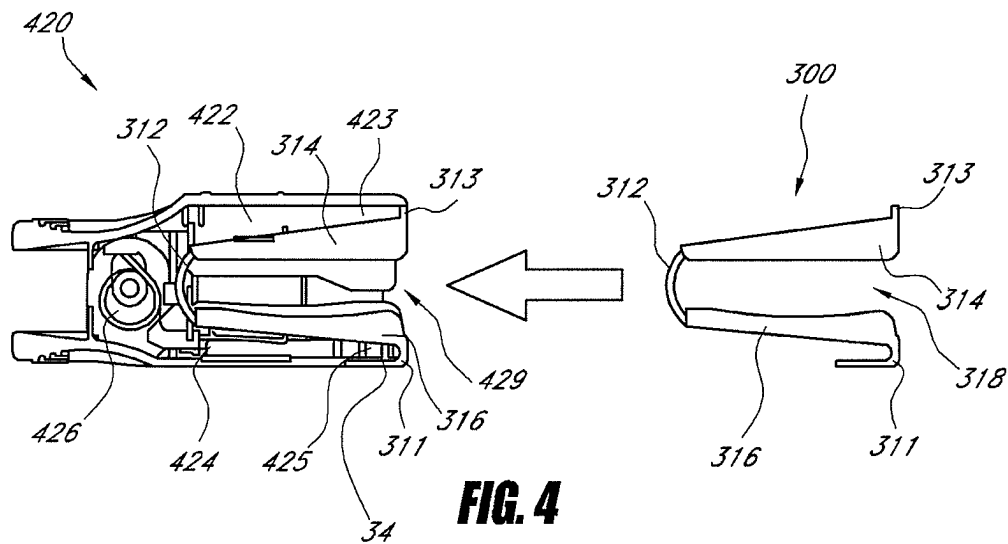
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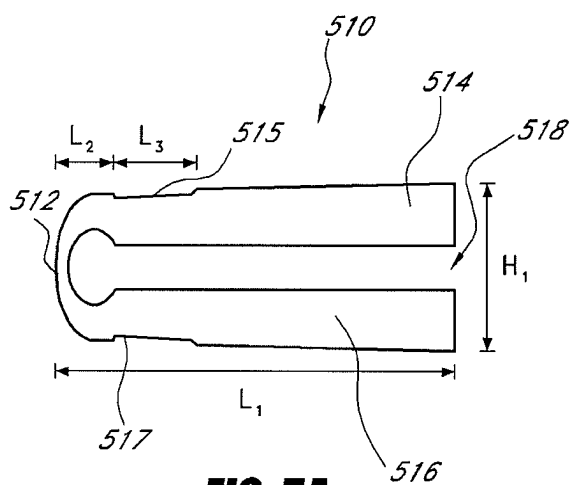


FIG. 5A

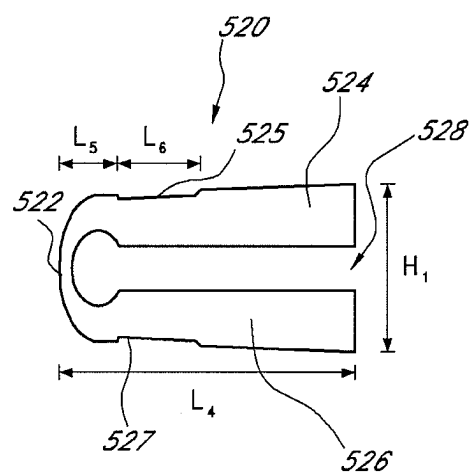
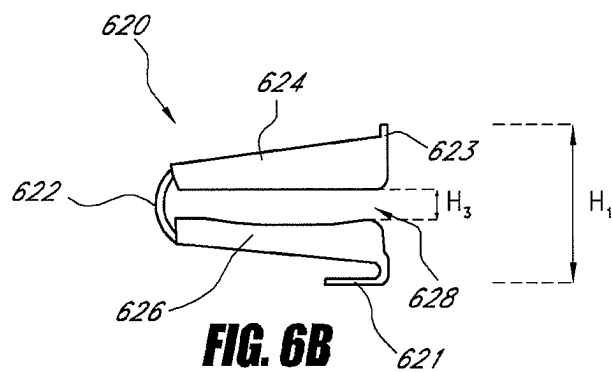
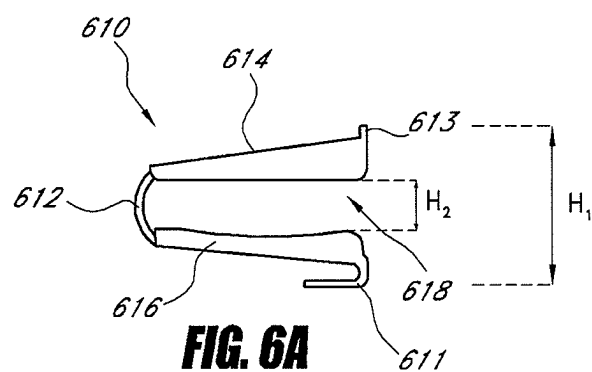


FIG. 5B



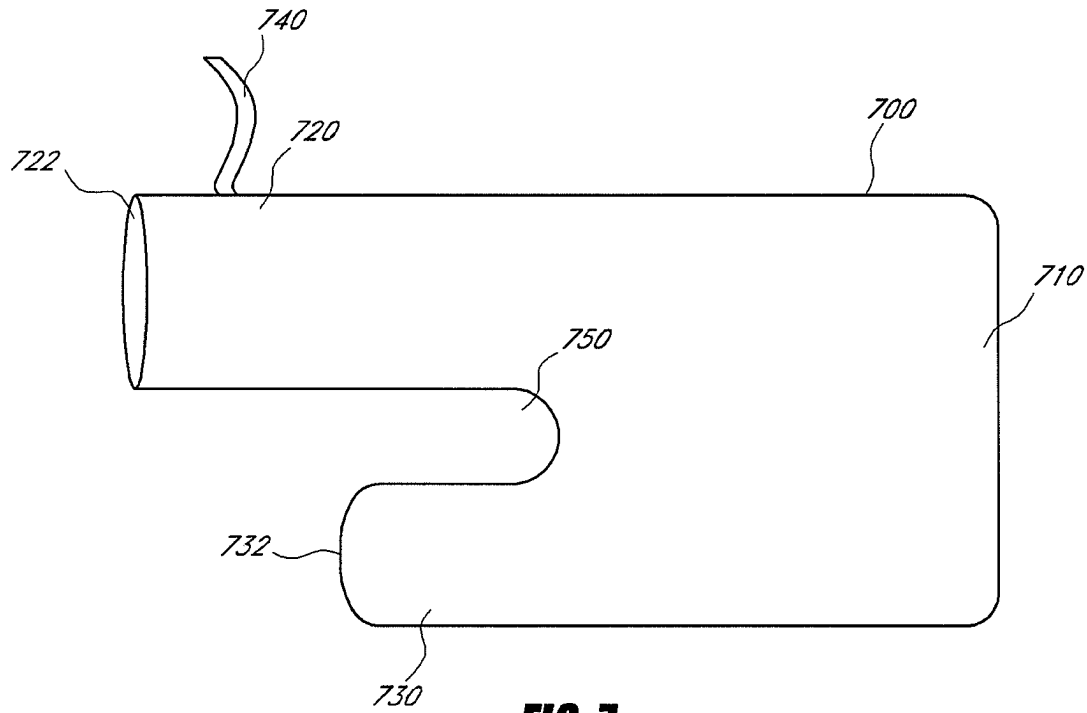


FIG. 7

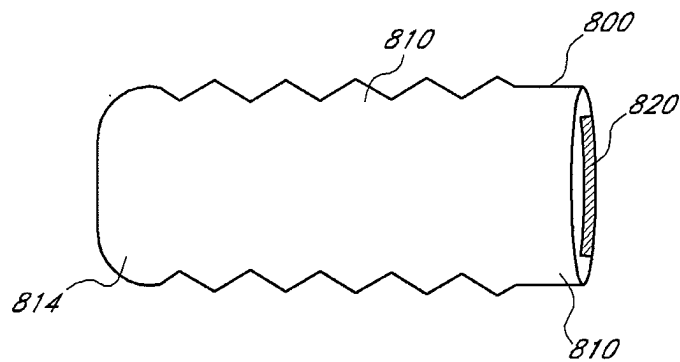


FIG. 8

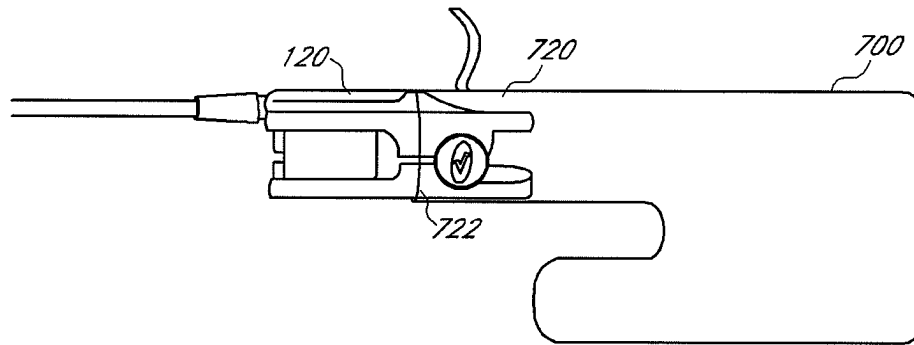


FIG. 9A

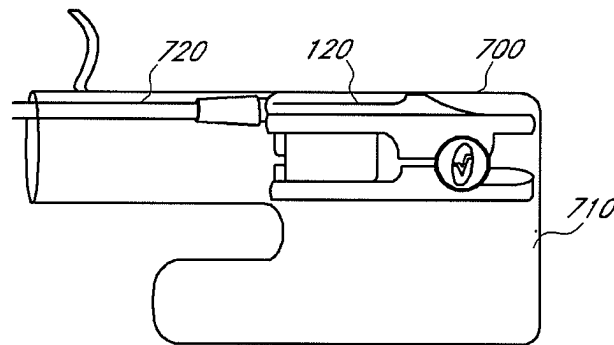


FIG. 9B

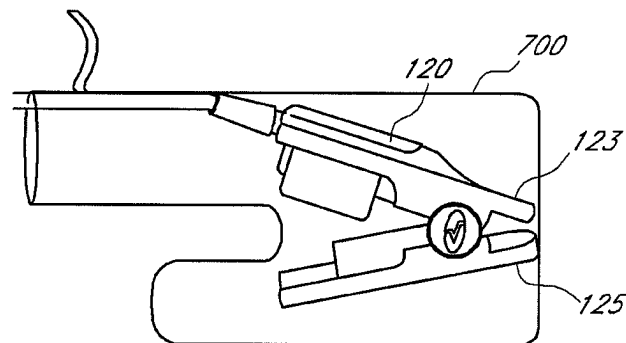


FIG. 9C

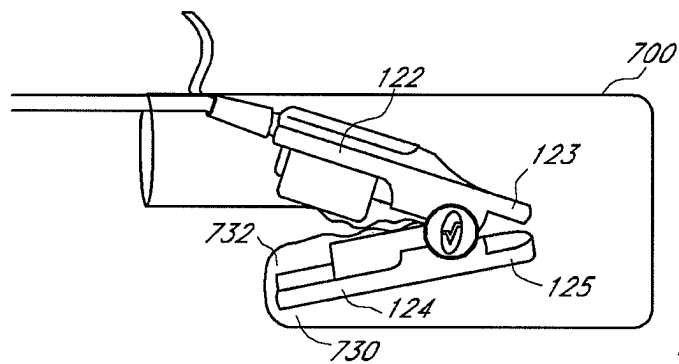


FIG. 9D

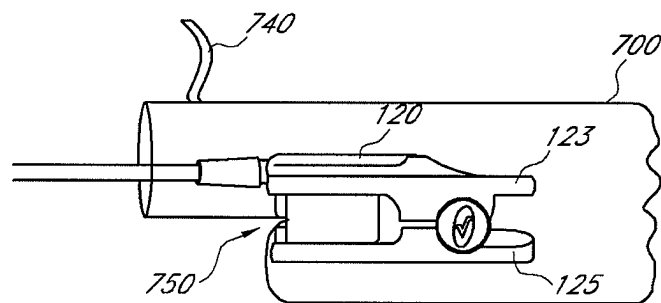


FIG. 9E

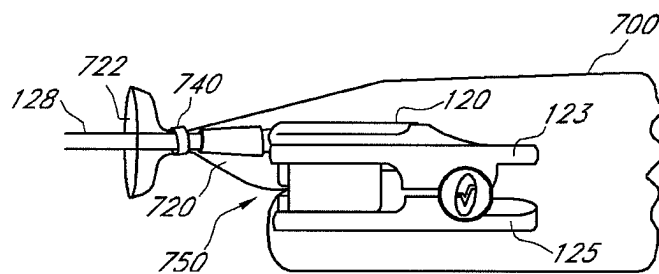


FIG. 9F

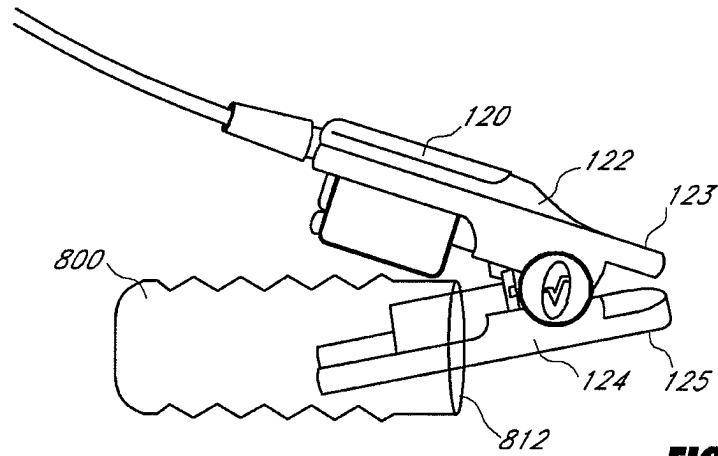


FIG. 10A

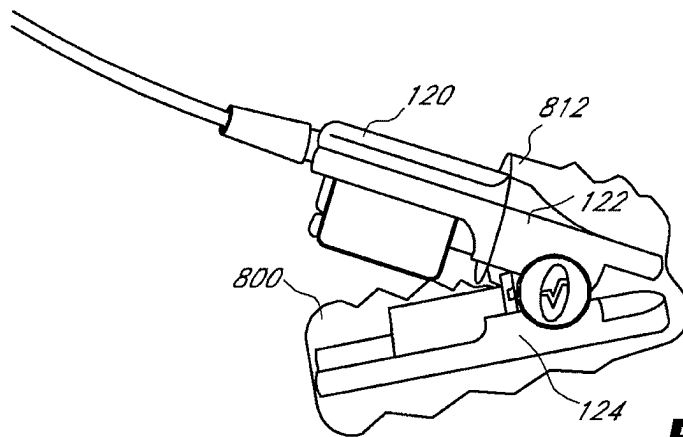


FIG. 10B

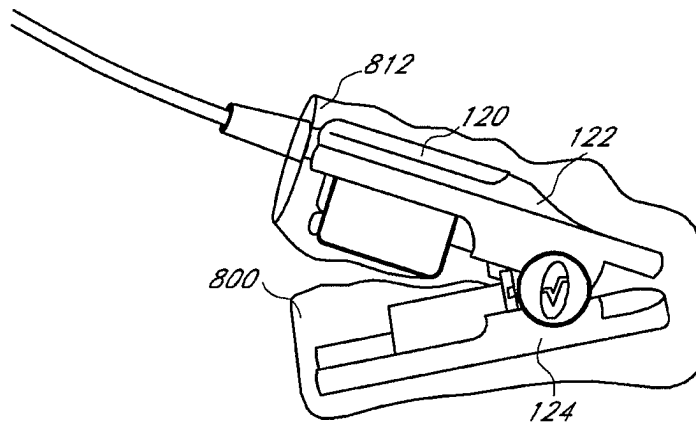


FIG. 10C

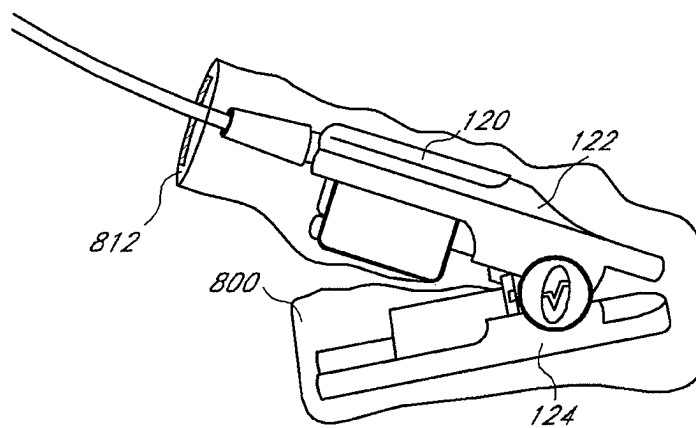


FIG. 10D

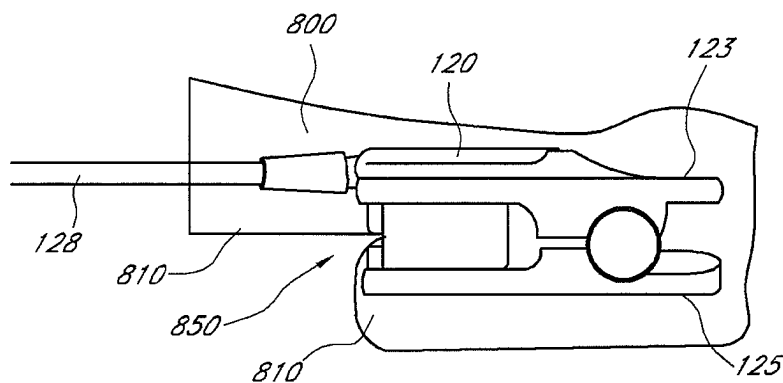


FIG. 10E

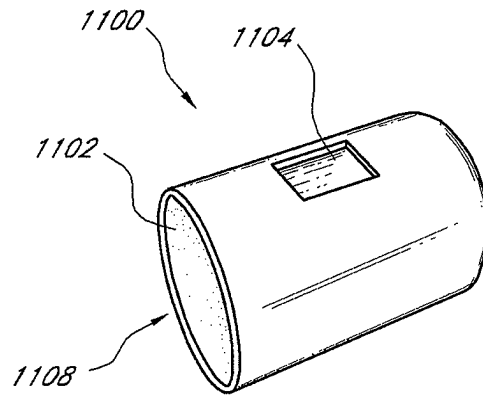


FIG. 11A

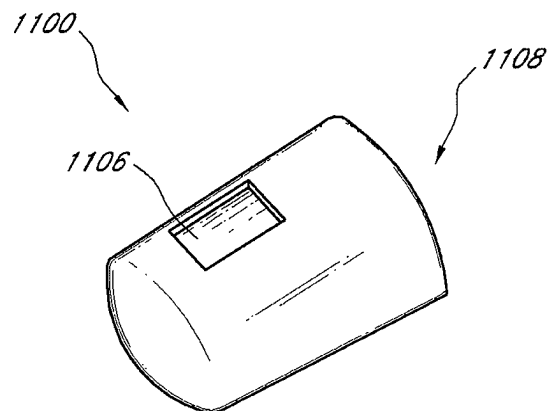


FIG. 11B

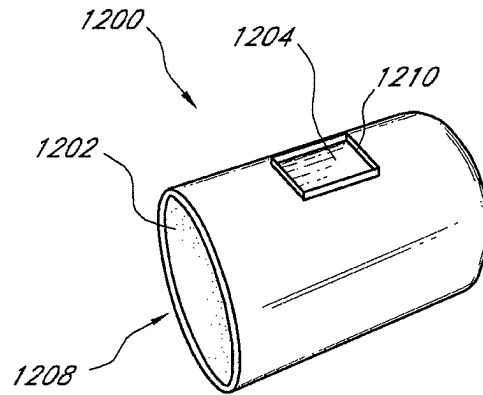


FIG. 12A

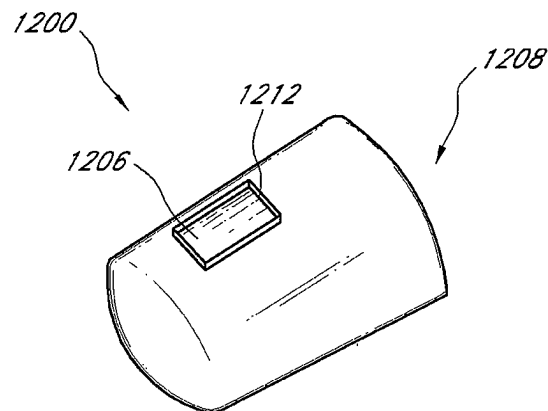


FIG. 12B

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**DISPOSABLE COMPONENTS FOR
REUSABLE PHYSIOLOGICAL SENSOR****CROSS-REFERENCE TO RELATED
APPLICATIONS**

The present application is a continuation of U.S. patent application Ser. No. 16/425,592, filed May 29, 2019, which application is a continuation of U.S. patent application Ser. No. 15/864,389, filed Jan. 8, 2018, which application is a continuation of U.S. patent application Ser. No. 14/626,570, filed Feb. 19, 2015, which application is a continuation of U.S. patent application Ser. No. 12/782,651, filed May 18, 2010, which application claims the benefit of priority from U.S. Provisional Application No. 61/179,670, filed May 19, 2009. The entire contents of each of the above items are hereby incorporated by reference herein for all purposes.

FIELD OF THE DISCLOSURE

The present disclosure generally relates to disposable components of non-invasive physiological sensors.

BACKGROUND OF THE DISCLOSURE

Non-invasive physiological sensors are applied to the body for monitoring or making measurements indicative of a patient's health. One application for a non-invasive physiological sensor is pulse oximetry, which provides a noninvasive procedure for measuring the oxygen status of circulating blood. Oximetry has gained rapid acceptance in a wide variety of medical applications, including surgical wards, intensive care and neonatal units, general wards, and home care and physical training. A pulse oximetry system generally includes a physiological sensor having light emitters and a detector, such as one or more LEDs and a light sensor. The sensor is attached to a tissue site, such as a finger, toe, ear lobe, nose, hand, foot, or other site having pulsatile blood flow which can be penetrated by light from the emitters. The detector is responsive to the emitted light after attenuation by pulsatile blood flowing in the tissue site. The detector outputs a detector signal to a monitor which processes the signal to provide a numerical readout of physiological parameters such as oxygen saturation (SpO₂) and pulse rate.

High fidelity pulse oximeters capable of reading through motion induced noise are disclosed in U.S. Pat. Nos. 6,770,028, 6,658,276, 6,157,850, 6,002,952, 5,769,785, and 5,758,644, which are assigned to Masimo Corporation ("Masimo") and are incorporated by reference herein. Advanced physiological monitoring systems may incorporate pulse oximetry in addition to advanced features for the calculation and display of other blood parameters, such as carboxyhemoglobin (HbCO), methemoglobin (HbMet) and total hemoglobin (Hbt), total Hematocrit (Hct), oxygen concentrations and glucose concentrations, as a few examples. Advanced physiological monitors and corresponding multiple wavelength optical sensors capable of measuring parameters in addition to SpO₂, such as HbCO, HbMet and Hbt are described in at least U.S. patent application Ser. No. 11/367,013, filed Mar. 1, 2006, titled Multiple Wavelength Sensor Emitters and U.S. patent application Ser. No. 11/366,208, filed Mar. 1, 2006, titled Noninvasive Multi-Parameter Patient Monitor, assigned to Masimo Laboratories, Inc. and incorporated by reference herein. Further, noninvasive blood parameter monitors and optical sensors including Rainbow™ adhesive and reusable sensors and RAD57™ and

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Radical-7™ monitors capable of measuring SpO₂, pulse rate, perfusion index (PI), signal quality (SiQ), pulse variability index (PVI), HbCO and HbMet, among other parameters, are also commercially available from Masimo.

SUMMARY OF THE DISCLOSURE

Optical sensors are widely used across clinical settings, such as operating rooms, emergency rooms, post anesthesia care units, critical care units, outpatient surgery and physiological labs, to name a few. Use in these settings exposes sensors, and in particular reusable sensors, to the potential risks of contamination and the resulting spread of nosocomial (hospital-acquired) infections. Studies have suggested that visual inspection of sensors may not detect contamination. Further, while a low-level disinfection protocol of alcohol wipes, dilute bleach scrubs or distilled water wipes can be effective when done correctly, reusable sensors may still be at risk for bacteria including MRSA (methicillin resistant *Staphylococcus aureus*). MRSA causes skin infections and can occasionally spread to almost any other organ in the body, sometimes with life-threatening potential. It is therefore a priority among medical care facilities to prevent contamination of reusable sensors by foreign and infectious materials and to prevent the spread of nosocomial infections. A sterile sensor cover advantageously provides a sterile environment without interfering with sensor functionality and capability. For example, the sterile sensor cover may be substantially impermeable to infectious agents (e.g., bacteria) and substantially optically transparent or transmissive.

Further, sensor elements such as the detector, emitters and associated circuitry can be expensive parts of a patient monitoring system. Often, relatively inexpensive degradable portions of the sensor become damaged due to repeated use or frequent sterilization. For example, the portions of the sensor which contact the user's skin often become soiled or damaged after each use. In such cases, users often disconnect the sensor cable from the monitor and replace the entire sensor. Moreover, in certain cases, such as for portable applications, it is useful for a sensor to be integrated into the patient monitor housing and not be attached by a cable. In such cases the sensor may not be removable from the monitor.

Other attempted solutions that include sensors with flexible, adhesive substrates having disposable and reusable portions. The disposable portion generally includes the adhesive portion of the sensor, which can lose its tack. Such sensors generally wrap around and adhesively attach to the tissue site. However, non-adhesive sensors having rigid housings, such as clip type sensors which clamp onto the tissue of the patient, are also commonly used in pulse oximetry and other patient monitoring applications. These sensors include degradable components that can become damaged or soiled due to frequent use, such as upper and lower pads which contact the user's skin.

It is therefore desirable to decouple the degradable portions from the rest of the sensor so that the degradable portions can be replaced in a cost-effective manner, and the rest of the sensor can be reused. A disposable sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor and protects the sensor from damage due to repeated use or sterilization.

In addition, incorrect alignment of a patient's tissue site with the sensor elements can lead to inaccurate results. For example, where the tissue site is a patient's finger, the emitter and detector should generally be aligned with the

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nail bed of the patient. Often it is difficult to determine whether the sensor is properly aligned because, for example, the sensor housing impedes the view of the tissue site in relation to the emitter and detector. In such cases, it may take the operator a significant amount of time to realize that a sensor is misaligned. As such, there is a need for a sensor which provides for robust tissue site alignment. Embodiments of the sensor cartridge are positionable on the user before placing the treatment site in the sensor, allowing for improved alignment of the treatment site.

In order to provide cost savings and allow for efficient use, it is also advantageous to be able to reuse sensors on different patients having tissue sites of various sizes, such as for both adults and children. The sensor cartridge of certain embodiments can be configured to allow a sensor to comfortably accommodate treatment sites of various sizes, such as for both adult and pediatric applications.

According to certain embodiments, a disposable sensor cartridge is provided for use with a noninvasive physiological sensor. The sensor cartridge can be capable of attaching to a tissue site and mating with a housing of the sensor. In certain embodiments, the sensor cartridge comprises a first portion comprising a first aperture configured to allow light emitted from one or more emitters of the sensor to travel through the first aperture such that the light is incident on a first region of a tissue site and travels through and is attenuated by body tissue of the tissue site. The attenuated light may exit the tissue site at a second region of the tissue site. In certain embodiments, the sensor cartridge also includes a second portion comprising a second aperture configured to allow the attenuated light to travel through the second aperture and to be received by a detector of the sensor. In some embodiments, the first and second portions coupled to define a cavity capable of receiving the tissue site. In certain embodiments, the sensor cartridge can also include an electrical component capable of electrical communication with the sensor. In some embodiments, the first portion is in contact with the first region while the cartridge is attached to the tissue site and the second portion is in contact with the second region while the cartridge is attached to the tissue site.

According to an aspect of the disclosure, a method of using a noninvasive physiological sensor having a housing is provided. The method can include providing a disposable sensor cartridge comprising a first portion comprising a first aperture and a second portion comprising a second aperture. In certain embodiments, the disposable sensor cartridge can also include an electrical component. The first portion and second portion may together define a cavity capable of receiving the tissue site. In certain embodiments, the method further includes attaching the sensor cartridge to the tissue site such that the tissue site is disposed within the cavity. The method according to some embodiments also includes mating the sensor cartridge with the sensor such that the sensor cartridge is disposed within a cavity of the sensor housing. In certain embodiments, mating of the sensor cartridge with the sensor is such that the electrical component is in electrical communication with a portion of the sensor.

In certain embodiments, a disposable sterile barrier is provided. The disposable sterile barrier may interpose a material impermeable to infectious biological substances between a tissue site and surfaces of a sensor configured to grasp the tissue site. The sensor may be configured to transmit optical radiation into the tissue site and to generate a signal responsive to the optical radiation after attenuation by pulsatile blood flow within the tissue site. In certain embodiments, the disposable sterile barrier comprises an

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optically transparent material that is substantially impermeable to infectious biological substances. The disposable barrier may comprise a first portion of the material formed such that the first portion can be interposed between a tissue site and a first surface of a reusable optical sensor proximate an emitter of the sensor. In certain embodiments, the disposable barrier may comprise a second portion of the material formed such that the second portion can be interposed between the tissue site and a second surface of the sensor proximate a detector of the sensor.

In certain embodiments, an optical sensing method of non-invasively measuring the constituents of pulsatile blood flow within a tissue site without substantial risk nosocomial infection by direct contact between a sensor and the tissue site. The optical sensing method may include providing a reusable optical sensor having an emitter disposed within a first housing and a detector disposed within a second housing, the emitter and detector in communication with a sensor cable, the first and second housings configured to be urged against opposite sides of a tissue site upon application of the sensor to the tissue site, the emitter configured to transmit optical radiation having one or more predetermined wavelengths into the tissue site and the detector configured to receive the optical radiation after attenuation by the tissue site. The sensor may be configured to generate one or more signals indicative of the attenuated radiation, the one or more signals transmitted via the sensor cable to a monitor configured to process the one or more signals to determine one or more physiological parameters of patient. The method may include applying a barrier so as to make physiological measurements without direct contact between the tissue site and the sensor. The barrier can be interposed between the tissue site and the first housing and interposed between the tissue site and the second housing upon application of the tissue site to the sensor, the barrier comprising material substantially impermeable to infectious substances and substantially transparent so as not to substantially distort the physiological measurements.

According to certain aspects of the disclosure, a disposable sterile barrier is provided comprising an elongate tube comprising a cavity. The elongate tube may comprise an open end and a closed end. In certain embodiments, the open end comprises an opening large enough to accommodate a reusable optical sensor. The cavity can be sized to fully enclose the reusable optical sensor and at least a portion of a sensor cable extending from the optical sensor. In some embodiments, the elongate tube comprises a material which is substantially optically transparent and substantially impermeable to infectious biological substances. In certain embodiments, the disposable sterile barrier may further include a fastener disposed proximate the open end of the elongate tube and configured to seal the open end of the elongate tube around the sensor cable.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a sensor cartridge and associated sensor according to an embodiment of the disclosure;

FIGS. 2A-D are side hinged-open, side hinged-closed, top and bottom perspective views, respectively, of the sensor cartridge of FIG. 1;

FIGS. 3A-D are side hinged-open and side hinged-closed views, top and bottom perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure;

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FIG. 4 illustrates the mating of the sensor cartridge of FIG. 3 with a sensor.

FIGS. 5A-B illustrate side views of two sensor cartridges of differing sizes according to an embodiment of the disclosure.

FIGS. 6A-B illustrate side views of two sensor cartridges of differing sizes according to another embodiment of the disclosure.

FIG. 7 is a side perspective view of a generally boot-shaped enclosure embodiment of a sterile sensor cover;

FIG. 8 is a side perspective view of a generally tube-shaped enclosure embodiment of a sterile sensor cover;

FIGS. 9A-F illustrate enclosing a reusable sensor with the boot-shaped enclosure and sealing the end with an attached tie; and

FIGS. 10A-E illustrate enclosing a reusable sensor with the tube-shaped enclosure and sealing the end with an adhesive strip.

FIGS. 11A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to an embodiment of the disclosure.

FIGS. 12A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure.

DETAILED DESCRIPTION

A sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor. Certain embodiments of the sensor cartridge protect the sensor from damage, such as damage due to repeated use, reduce the need for sensor sanitization, or both. Further, embodiments of the sensor cartridge are positionable on the user before insertion in the sensor and allow for improved alignment of the treatment site with the sensor. In addition, the sensor cartridge of certain embodiments of the disclosure can be configured to allow a single sensor to comfortably accommodate treatment sites of various sizes such as for both adult and pediatric applications. The terms "sensor cover" and "sensor cartridge" are used throughout to describe various embodiments of the disclosure. The terms may be used interchangeably and are not intended to be limiting.

The tissue site of the illustrated embodiments is a finger and the following description therefore refers specifically to the tissue site as a finger for the purposes of clarity. This is not intended to be limiting and, as described herein, the sensor cartridge 110 of certain embodiments may be used with other types of tissue sites.

FIG. 1 is a perspective view of a sensor cartridge 110 and associated sensor 120 according to an embodiment of the disclosure. A user can place a cartridge 110 on a finger of the patient and then insert the finger along with the attached cartridge 110 into a non-invasive physiological sensor 120.

The sensor 120 can be a clip-type sensor including an upper housing 122, a lower housing 124 and a hinge element 126. The upper and lower housings 122, 124 house electrical and/or optical components (not shown) of the non-invasive physiological sensor 120. For example, the upper and lower housings 122, 124 may house light emitters and a detector of a pulse oximeter sensor, such as one or more LEDs and a light sensor. The sensor 120 can be connected to a patient monitor (not shown) via a cable 128. For example, the detector outputs a signal to the monitor over the cable 128 which then processes the signal to provide a numerical readout of physiological parameters such as oxygen saturation (SpO₂) and pulse rate.

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After placing the sensor cartridge 110 on the finger, the user can attach the sensor 120 to the patient by applying pressure to the ends 123, 125 of the upper and lower housings 122, 124, forming a cavity 129 capable of receiving the tissue site. Once the patient inserts the tissue site along with the attached sensor cartridge 110 into the cavity 129, the pressure on the ends 123, 125 can be released such that the upper and lower housings 122, 124 come in contact with and secure the tissue site, allowing for accurate non-invasive physiological measurement.

Although disclosed with reference to the sensor of FIG. 1, an artisan will recognize from the disclosure herein a wide variety of oximeter sensors, optical sensors, noninvasive sensors, medical sensors, or the like that may benefit from the sensor cartridges disclosed herein. The sensor may not be a hinge type sensor but may instead, for example, include an integral housing having an annular ring type opening through which the patient inserts their finger. In various embodiments, the sensor may be adapted to receive a tissue site other than a finger such as a, toe, ear lobe, nose, hand, foot, or other site having pulsatile blood flow which can be penetrated by light from the emitter. In addition, the cartridge 110 may be used with a portable monitor and associated sensor components in certain embodiments. Such monitors, including the sensor components, can be integrated into a hand-held device such as a PDA and typically do not include cables or separate monitors. Portable monitors are often used by first responders in emergency situations, in part because of their portability and ease of use. As such, disposable cartridges 110 which can protect the sensor components according to embodiments herein can be of particular benefit when used with spot-check monitors. In addition, in emergency situations medical personnel must treat a large number of patients relatively quickly and it may therefore be difficult to sanitize sensors between patients. Disposable cartridges 110 described herein benefit medical personnel in such cases because a separate pre-sterilized cartridge 110 can be used for each patient.

Referring still to FIG. 1, the cartridge 110 includes an upper portion 114, a lower portion 116 and a hinge portion 112 which together form a cavity 118 capable of receiving a finger. The upper portion 114, lower portion 116 and hinge portion 112 are generally curved so as to accommodate the natural shape of the finger. A hinge cut-out 113 can be disposed on either side of the cartridge 110 near the hinge portion 112, allowing for the upper and lower portions 114, 116 to separate from each other by rotating about the hinge portion 112. To apply the cartridge to the finger, the upper portion 114 and lower portion 116 may be separated such that they rotate about the hinge portion 112 and the cavity 118 becomes large enough to comfortably receive the finger. Once the finger is placed in the cavity 118, the upper and lower portions 114, 116 are released, coming into contact with and releasably attaching to the finger. For example, in certain embodiments, the hinge portion 112 may provide a spring force which is transferred to the upper and lower portions 114, 116 when they are released such that they grab onto the finger. In some embodiments, the interior surface 119 of the cartridge 110 may include an adhesive substance such that the cartridge 110 is releasably attached to the finger or other tissue site. In other configurations, the cartridge 110 simply rests on the finger and there is not a separate attachment mechanism.

The upper portion 114 includes an upper aperture 115 and the lower portion 116 includes a lower aperture 117. The apertures 115, 117 generally allow for proper sensor operation. For example, the apertures 115, 117 allow for light

from one or more emitters of the sensor 120 to contact the finger and for light attenuated by the tissue site to be received by a detector of the sensor 120.

Because the upper and lower sensor housings 122, 124 are often opaque, it can be difficult to determine whether or not a finger is properly aligned once it is placed in the sensor. The apertures 115, 117 can allow for proper alignment of the sensor cartridge 110 with respect to the finger prior to inserting the cartridge 110 and finger into the sensor 120. Proper alignment of the cartridge 110 with the finger can therefore improve the accuracy of measurements by helping to ensure that the finger will be properly aligned with respect to the sensor 120 elements upon mating of the cartridge 110 and sensor 120. In certain cases, medical personnel may not realize that an inaccurate measurement has occurred, or may not realize it until after an alarm on the sensor goes off, after removal of the sensor or after the patient has left the treatment facility. As such, proper alignment can also help reduce cost and save time.

The upper aperture 115 allows a user to visually determine whether the nail bed of a finger is properly aligned with the aperture 115. Proper alignment of the nail bed also ensures that the lower aperture 117 is properly aligned with the fleshy part of the underside of the finger. In addition, the user may visually determine that the underside of the finger is properly aligned by looking through the lower aperture 117. The finger with the properly aligned cartridge 110 can then be inserted into the sensor 120 such that the light from the emitters of the sensor 120 will be incident on the nail bed through the upper aperture 115. Further, the detector of the sensor 120 then receives attenuated light from the appropriate portion of the underside of the finger through the lower aperture 117. In other embodiments, the emitter of the sensor 120 is housed in the lower housing 124 and the detector is housed in the upper housing 122.

In certain embodiments, a film or other material (not shown) may be included in one or more of the apertures 115, 117. For example, in one embodiment, translucent plastic film is placed in the apertures 115, 117 such that the properties of the emitted and attenuated light passing through the apertures are not affected by the material. In other embodiments, material is used which does affect the optical properties. For example, an optical film which filters out particular wavelengths of light is used in some embodiments.

Tactile feedback elements (not shown) may indicate proper alignment of the cartridge instead of, or in addition to, the apertures 115, 117. For example, one or more of the upper and lower portions 114, 116 may include small protrusions which indicate to the patient whether the cartridge 110 is properly aligned. In one embodiment, for example, protrusions on either side of the finger indicate to the patient that the cartridge 110 is centered on the finger and a protrusion near the hinge portion 112 indicates that the finger is inserted appropriately deep into the cartridge 110. Other tactile elements such as, for example, recesses, may be used in other embodiments.

In certain embodiments, the properties of the interior 119 of the cartridge 110 allow for proper and efficient calibration of the sensor 120. For example, the interior 119 of the cartridge may be a highly reflective color such as white. The interior 119 may also have a glossy texture in certain embodiments, which can also aid in the calibration of the sensor.

As discussed, the sensor cartridge 110 is configured to mate with the sensor 120. For example, the cartridge 110 of FIG. 1 mates in a friction fit with the sensor 120. To release

the cartridge 110 from the sensor 120 after use, the user opens the sensor 120 clip, thereby relieving the pressure on the cartridge and tissue site from the upper and lower housings 122, 124. The user can then readily pull the tissue site and cartridge 110 out of the cavity 129. As will be appreciated by skilled artisans from the disclosure provided herein, other types of mating and release mechanisms are possible. For example, snap-fit mating configurations such as the one described below with respect to FIG. 3 can be used. In some embodiments, a male or female track on the cartridge 110 fits into a corresponding male or female track on the sensor 120, as appropriate. A button or lever release mechanism may be used in certain embodiments. In some embodiments, the cartridge 110 may automatically release when the sensor 120 clip is opened.

The sensor cartridge 110 or a portion thereof can be constructed of urethane rubber plastic in certain embodiments. In various other configurations, as will be appreciated from the disclosure herein, the cartridge 110 may be made of other appropriate materials such as Acrylonitrile Butadiene Styrene ("ABS") or other types of rubber or plastic materials. The cartridge 110 is formed with biodegradable material in certain embodiments.

The general structure of the sensor cartridge 110 may differ in various configurations. For example, the cartridge may not be an integral molded piece as shown in the embodiment of FIG. 1 and may instead include separate pieces as discussed with respect to FIG. 3 below.

In an example scenario, once the user attaches the properly aligned sensor cartridge 110 to the tissue site and applies the sensor 120 as discussed above, medical personnel take a measurement of one or more physiological parameter of the patient using the sensor 120. The patient then removes the tissue site from the sensor 120 along with the attached cartridge 110 as discussed and the patient or medical personnel may dispose of the cartridge 110. Those of skill in the art will recognize from the disclosure herein various ways of using the cartridge 110. For example, the cartridge 110 of certain embodiments may be mated with the sensor 120 before application to the tissue site. Moreover, the cartridge may be used more than once. For example, the cartridge 110 may be used several times by the same patient before disposal. In some embodiments, the cartridge 110 may be sterilized between uses.

In certain embodiments, the sensor cartridge 110 is configured to be in electrical communication with the sensor 120. For example, the sensor cartridge 110 can also include one or more electronic components 130 and one or more coupling portions (not shown) which may be electrically coupled to the sensor 120. For example, the coupling portions may comprise one or more electrical contacts (e.g., solder pads), electrical connectors (e.g., a socket and pin type connector), and the like. The electronic components 130 may include one or more information elements in certain embodiments. The information element may comprise one or more memory devices, such as, for example, one or more EPROMs or EEPROMs (e.g., those commercially available from Dallas Semiconductor), other memory or processing devices, combinations of the same, or the like. In some embodiments, the information element includes a conductor, a resistor, a single wire addressable memory device, or the like. In general, the information element may include a read-only device or a read and write device. The information element may advantageously comprise a resistor, an active network, or any combination of the foregoing. The information element may include data accessible by the sensor 120 and/or attached patient monitor to accomplish

quality control, use monitoring, combinations of the same, or the like. For example, the information element may provide identification information to the system (e.g., the sensor 120 and/or monitor) which the system uses to determine whether the cartridge is compatible with the system. In an advantageous embodiment, the monitor reads the information element to determine one, some or all of a wide variety of data and information, including, for example, a type of patient, cartridge manufacturer information, life data indicating whether the cartridge has been used and/or should be replaced, encryption information, keys, indexes to keys or has functions, or the like monitor or algorithm upgrade instructions or data, some or all of parameter equations, information about the patient, age, sex, medications, and other information that can be useful for the accuracy or alarm settings and sensitivities, trend history, alarm history, sensor life, or the like.

The information element may be positioned on the hinge portion 112. For example, in one embodiment the information element be embedded in the cartridge 112 material of the hinge portion 112 and is electrically coupled via a connector extending from the hinge portion 112 which mates with a corresponding connector on the interior of the sensor 120. Skilled artisans will recognize from the disclosure provided herein a variety of configurations for the placement of the information element. For example, in various embodiments, the information element may be located on one or more of the inner surface of the cartridge 110, the outer surface of the cartridge 110, or embedded within the cartridge 110 material. Moreover, the information element may be positioned on the hinge portion 112, the upper portion 114, the lower portion 116, or a combination thereof.

The information element may advantageously store some or all of a wide variety of data and information, including, for example, information on the type or operation of the sensor cartridge, buyer or manufacturer information, software such as scripts, executable code, or the like, sensor cartridge 110 life data indicating whether the sensor cartridge 110 has expired and should be replaced, encryption information, etc. or monitor or algorithm upgrade instructions or data. In various configurations, the information element may advantageously configure or activate the monitor, monitor algorithms, monitor functionality, or the like based on some or all of the foregoing information. For example, without authorized data accessible on the information element, quality control functions may inhibit functionality of the monitor. Likewise, particular data may activate certain functions while keeping others inactive. For example, a particular cartridge 110 may be compatible for use in measuring one type of physiological parameter of a set of physiological parameters the monitor is capable of measuring. In such a circumstance, the monitor may only activate measurement of the one type of physiological parameter based on the data accessible from the information element. Further information regarding information elements and systems and methods for monitoring sensor life can be found in U.S. Publication No. 2008/0088467, which is hereby incorporated in its entirety by reference herein.

While disclosed with respect to the cartridge 110 of FIGS. 1-2, any of the cartridges described herein, such as the cartridges 300, 510, 520, 610, 620, 1100, 1200 described below of FIGS. 3-6 and FIGS. 11-12 may include one or more information elements. Likewise, any of the sensor covers disclosed herein, such as the sensor covers 700, 800 of FIGS. 7-10 may include an information element.

FIGS. 2A-D are side hinged-open, side hinged-closed, top-right perspective and top-left perspective views, respec-

tively, of the sensor cartridge of FIG. 1. As discussed above, the upper portion 114 and lower portion 116 may be separated such that they rotate about the hinge portion 112 and the cavity 118 becomes large enough to comfortably receive the finger (FIG. 2A). Once the finger is placed in the cavity 118, the upper and lower portions 114, 116 can be released, coming into contact with and releasably attaching to the finger (FIG. 2B). The right and left sides of the cartridge 110 of the illustrated embodiment are symmetrical. As shown by FIGS. 2C-D, the top and bottom are also symmetrical. In other embodiments, the left and right sides and/or the top and bottom may be shaped differently, such as, for example, to accommodate asymmetric properties of the finger or other tissue site.

FIGS. 3A-D are side hinged-open and side hinged-closed views, top-right perspective and top-left perspective views, respectively, of a sensor cartridge 300 according to another embodiment of the disclosure. The cartridge 300 includes an upper portion 314, a lower portion 316 and a hinge portion 312, which are three separate pieces. As shown, the hinge portion 312 attaches to the back 302 of the upper portion 314 and the back 304 of the lower portion 316, forming a cavity 318 capable of receiving a finger of a patient. The hinge portion 312 may attach to the upper and lower portions 312, 314 with glue, may be formed integrally with the upper and lower portions 312, 314, or may be connected in some other manner. As shown with respect to FIGS. 3C-D, the cartridge 110 further includes upper and lower apertures 315, 317, which may be similar in structure and function to the apertures 115, 117 described above with respect to FIGS. 1-2.

FIG. 4 illustrates the mating of the sensor cartridge of FIG. 3 with a sensor 420 (shown in a cross-sectional view). The sensor 420 may be similar in structure and function to the sensor 120 of FIG. 1, for example, and can include an upper housing 422, a lower housing 424 and a hinge portion 426 which form a cavity 429. One or more securing features may be included on the cartridge 300 to secure the cartridge 300 with the one or more corresponding securing features on the sensor 420 upon mating to secure the cartridge to the sensor 420 and/or ensure the correct position of the cartridge 300 within the sensor 420. Referring to FIGS. 3 and 4, for example, the attachment arm 311 is configured to secure the cartridge 300 in place when applied to the sensor 420. Upon application to a sensor 420 the front portion 425 of the lower housing 424 of the sensor 420 may occupy the space defined by the attachment arm 311 and the underside of the lower portion 316 of the cartridge 300. The attachment arm 311 helps to releasably secure the sensor 420, via a friction fit, for example. One or more other features, such as the lip 313 disposed on the upper portion 314 of the cartridge 300 can be included to further secure the cartridge 300 in the sensor 420. Upon insertion of the cartridge 300 into the sensor 420, the front portion 423 of the upper housing 422 abuts the lip 313. Accordingly, the lip 313 can help ensure that the cartridge 300 is positioned appropriately deep within the sensor 420.

The shape of the cartridge 300 may be configured to secure the cartridge 300 and/or ensure proper positioning of the cartridge 300 upon mating with the sensor 420. For example, as shown in FIGS. 3-4, the upper portion 314 and/or the lower portion 316 can be sloped such that they thicken towards the front 301 of the cartridge 300. Upon insertion, the sloping feature can provide a friction fit with the corresponding portions of the sensor 420.

As will be appreciated by skilled artisans from the disclosure provided herein, various attachment and positioning

mechanisms may be used. For example, the attachment arm **311** may further include a protrusion or other feature which may fit into a corresponding feature, such as a recess (not shown), on the underside of the lower housing **424** of the sensor **420**. In other configurations, the features may be reversed. For example, the protrusion may be on the lower housing **424** and the recess may be included on the attachment arm **311** of the cartridge **300**.

FIGS. 5A-B illustrate side views of two sensor cartridges **510**, **520** of differing sizes according to embodiments of the disclosure. The sensor cartridges **510**, **520** may be similar in structure and function to the cartridge **100** of FIGS. 1-2 and include upper portions **514**, **524**, lower portions **516**, **526**, hinge portions **512**, **522**, cavities **518**, **528**, upper apertures **515**, **525** and lower apertures **517**, **527**, respectively. The cartridges **510**, **520** are configured to accommodate treatment sites of various sizes, such as for both adult and pediatric applications. For example, the length L_1 of the cartridge **510** of FIG. 5A is larger than the corresponding length L_4 of the cartridge **520** of FIG. 5B. Additionally, the upper and lower apertures **515**, **517** of the cartridge **510** of FIG. 5A are set back a length L_2 from the rear of the cartridge **510** which is relatively large in comparison to the length L_5 that the upper and lower apertures **525**, **527** of cartridge **520** of FIG. 5B are set back from the rear of the cartridge **520**. The length L_3 of the upper and lower apertures **515**, **517** of the cartridge **510** of FIG. 5A is relatively large in comparison to the length L_6 of the upper and lower apertures **525**, **527** of the cartridge **520** of FIG. 5B.

The cartridge **510** is configured accommodate a larger finger than the cartridge **520**. For example, the cartridge **510** may be used in adult applications and the cartridge **520** may be used in pediatric applications. A pediatric patient, for example, may have a nail bed that is set back a relatively short distance from the tip of the finger in comparison to an adult patient. As such, the shorter set back length L_5 may be appropriate for a pediatric patient while the length L_2 may be appropriate for an adult patient. A pediatric patient may also have a relatively smaller nail bed than an adult patient such that the smaller length L_6 appropriately accommodates a pediatric patient while the length L_3 may more appropriately accommodate an adult patient. Finally, pediatric patients typically have shorter fingers than adult patients such that the shorter overall cartridge length L_4 may be appropriate for a pediatric patient while the longer length L_1 may be more appropriate for adult patients.

As shown, the external profile of the cartridges **520**, **521**, characterized in part by the maximum height of the cartridges **520**, **521** H_1 may be the same. As such, the cartridges **520**, **521** may mate with a single sized sensor, thereby reducing cost and complexity. One of skill in the art will recognize from the disclosure herein other configurations. Some of the lengths which are shown as different in the embodiments **510**, **520** of FIGS. 5A-B may be the same in alternative configurations. For example, in various embodiments, one or more of the pairs of lengths L_1 and L_4 , L_2 and L_5 , L_3 and L_6 may be the same.

FIGS. 6A-B illustrate side views of two sensor cartridges **610**, **620** of differing sizes according to embodiments of the disclosure. The sensor cartridges **610**, **620** may be similar in structure and function to the cartridge **300** of FIG. 3 and include upper portions **614**, **624**, lower portions **616**, **626**, hinge portions **612**, **622**, cavities **618**, **628**, attachment arms **611**, **621** and lips **613**, **623** respectively. The cartridges **610**, **620** are configured to accommodate treatment sites of various sizes, such as for both adult and pediatric

applications. For example, the upper and lower portions **614**, **616** of the cartridge **610** are thinner than the corresponding portions **624**, **626** of the cartridge **620** and define a relatively large cavity **618**, characterized in part by the length H_2 . On the other hand, the upper and lower portions **624**, **626** of the cartridge **620** are relatively thick and define a relatively small cavity **628**, characterized in part by the length H_3 . As such, the cartridge **610** is configured accommodate a larger finger than the cartridge **620**. For example, the cartridge **610** may be used in adult applications and the cartridge **620** may be used in pediatric applications.

As shown, the external profile of the cartridges **620**, **621**, characterized in part by the maximum height of the cartridges **620**, **621** H_1 may be the same. As such, the cartridges **620**, **621** may mate with a single sized sensor, thereby reducing cost and complexity. One of skill in the art will recognize from the disclosure herein other configurations. In some embodiments, the hinge portions **612**, **614** are also shaped to accommodate tissue sites of various sizes. In some configurations, the upper portion, the lower portion and/or the hinge portion of the sensor cartridge are constructed of a resilient material (e.g., a spongy material) which conforms to the shape of the finger such that a sensor cartridge of a single size can accommodate tissue sites of multiple sizes. In some embodiments in which the opening of the cartridge is generally circular, the circumference of the opening through which the patient inserts their finger into the cavity **518** is larger than the opening of the cartridge **528**.

FIGS. 7-8 illustrate two embodiments of a sterile cover **700**, **800** each advantageously configured to enclose a reusable sensor **120** (FIG. 1). In this manner, a patient's tissue site, such as a finger, can be inserted into the sensor and a physiological measurement made without contamination of the sensor or the patient. The sensor and the patient are thus advantageously protected from exposure to infectious agents such as MRSA.

As shown in FIG. 7, a boot-shaped cover **700** defines an enclosure having a generally tubular leg **720** and a smaller, generally tubular foot **730** both extending from a cross-member **710**. In some embodiments, the cross-member **710** is sealed or substantially sealed. The leg **720** has an open end **722** of sufficient diameter to receive a reusable sensor **120**, as described with respect to FIGS. 9A-B, below. The foot **730** has a closed end **732** of sufficient diameter to receive the lower housing **124** of a reusable sensor **120**, as described with respect to FIGS. 9C-D, below. A tie **740** wraps around the open end **722** so as to close that end around the sensor cable **128**. In one embodiment, the tie **740** is a self-adhesive tie, for example. In an embodiment, the tie **740** wraps around the open end **722** so as to seal the end around the sensor cable **128**. A gap **750** between the leg **720** and foot **730** of the cover **700** defines a space configured to accommodate a finger-tip within the sensor **120**.

As shown in FIG. 8, a tube-shaped cover **800** defines an enclosure having a generally cylindrical body **810** with an open end **812** and a closed end **814**. The open end **812** is of sufficient diameter to receive first the lower housing **124** followed by the upper housing **122** of a reusable sensor **120**, as described with respect to FIGS. 10A-D, below. A self-adhesive strip **820** disposed internal to the body **810** proximate the open end **812** closes the open end **812** around the sensor cable **128**. In one embodiment, for example, the open end **812** forms a seal around the sensor cable **128**. In an embodiment, the covers **700**, **800** include of an optically transparent or substantially transparent material so as to negligibly attenuate or otherwise negligibly distort those wavelengths in the red and infrared spectrum utilized in

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pulse oximeters or otherwise for the measurement of blood constituents. For example, the covers **700**, **800** may be substantially optically transmissive.

As used herein, the term “tube” is used to describe a generally elongate member comprising a cavity and may include an open end and a closed end, for example. As used herein, a tube may have a variety of shapes and characteristics. For example, the tube **800** of various may comprise a cylindrical, rectangular, ovalar, triangular, or some other cross-sectional shape, or may be generally deformable.

FIGS. **9A-F** illustrate the use of a sterile cover **700** embodiment to enclose a reusable sensor **120**, thereby advantageously protecting the sensor from contamination and the patient from exposure to infectious agents. A reusable sensor **120** in its normally closed position is initially inserted into the open end **722** of the cover **700** (FIG. **9A**). The sensor **120** is then pushed through the length of the leg **720** until, for example, the sensor abuts the wall of the cross-member **710** (FIG. **9B**). The sensor grips **123**, **125** are pressed from outside the cover **700** so as to move the sensor **120** to its open position (FIG. **9C**). While maintaining pressure on the grips **123**, **125**, the sensor lower housing **124** slides into the foot **730** until it abuts the closed end **732** (FIG. **9D**). Pressure is released from the grips **123**, **125** so that the sensor **120** returns to its closed position over the gap **750** (FIG. **9E**). The tie **740** is wrapped around the leg **720** so as to close-off the end **722** tightly around the sensor cable **128** (FIG. **9F**). The covered sensor **120** is now ready for single-patient use by squeezing the grips **123**, **125** to open the sensor and by inserting a patient finger into the cover gap **750**, which is now disposed inside the sensor **120**, between the emitters and detector.

FIGS. **10A-E** illustrate the use of another sterile cover **800** embodiment to enclose a reusable sensor **120**, likewise substantially protecting the sensor and patient from contamination and/or infectious agents. Sensor grips **123**, **125** are pressed to place the sensor **120** in an open position (FIG. **10A**). The cover **800** may be initially in an “accordion” state or otherwise folded, rolled or compressed, or otherwise collapsible to a reduced overall length. The cover opening **812** is slipped over the lower housing **124** (FIG. **10A**). Maintaining the sensor **120** in the open position, the cover **800** is unfolded or otherwise extended in length as the opening **812** is pulled over the grips **123**, **125** and onto the upper housing **122** (FIG. **10B**). The cover opening **812** is pulled further along the upper housing **122** and onto the sensor cable **128** (FIGS. **10C-D**). The sensor grips **123**, **125** are released so that the sensor **120** returns to its normally closed position (FIG. **10E**). The self-adhesive strip **820** just inside the opening **812** is then exposed, such as by peeling back and removing a liner, and the opening **812** is securely closed around the cable **128** (FIG. **10E**). The covered sensor **120** is now ready for single-patient use by squeezing the grips **123**, **125** to open the sensor and by inserting **850** a patient finger into the sensor between the folded over wall **810** portions of the cover **800** (FIG. **10E**). This places the patient’s fingertip inside the sensor **120**, between the emitters and detector.

Skilled artisans will recognize a variety of alternatives sensor covers **700**, **800** from the disclosure provided herein. For example, in other embodiments, the covers **700**, **800** may include one or more apertures such as one or more of the apertures described herein. The covers **700**, **800** may comprise a rigid or semi-rigid material. Moreover, the covers **700**, **800** may be used in addition to a cartridge such as one or more of the cartridges described herein with respect to FIGS. **1-6** and **11-12**. In one alternative embodi-

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ment, the cover **700**, **800** may extend beyond the sensor **120** along some length of the cable **128**. For example, the cover **700**, **800** may extend substantially along the entire cable to the monitor in one embodiment. In another embodiment, the cover **700**, **800** may extend partially along the cable **128**.

FIGS. **11A-B** illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge **1100** according to an embodiment of the disclosure. In certain circumstances, light incident on the finger from the emitter may not be entirely absorbed by the finger. In addition, a certain portion of attenuated light will exit the other side of the finger. Such portions of light may reflect within the space between the cartridge and the finger or between the cartridge and the sensor. These portions of light can interfere with each other, with the emitted light and/or with the attenuated light. These portions of reflected light and associated interfering light can be incident on the detector. This phenomena, sometimes referred to herein as “light bounce,” can potentially cause inaccurate measurements, depending on certain factors such as the amount of light bounce and the relative sensitivity of the measured signal.

The sensor cartridge **1100** is configured for application to a tissue site such as a finger, for example. The finger or other tissue site may be placed in the opening **1102**. In addition, the cartridge **1100** is configured for insertion into a reusable sensor such as the sensor **120**. The sensor cartridge **1100** further includes an upper aperture **1104** and a lower aperture **1106**. The apertures **1104**, **1106** generally allow for proper sensor operation. For example, the apertures **1104**, **1106** allow for light from one or more emitters of the sensor **120** to contact the finger and for light attenuated by the tissue site to be received by a detector of the sensor **120**. The apertures **1104**, **1106** may function in a manner similar to and provide similar advantages as the apertures **115**, **117** of the cartridge **110** of FIGS. **1-2**. The cartridge may comprise Acrylonitrile Butadiene Styrene (“ABS”), other types of rubber or plastic materials, or some other material compatible with the embodiments described herein.

As shown, the sensor cartridge **1100** generally envelopes the finger when it is applied to the cartridge **1100**. Moreover, the cartridge **1100** may comprise a color absorptive of the light emitted from the emitter of the sensor **120**. For example, the cartridge **1100** may comprise a dark material as shown, such as a substantially black or opaque material. As described above, a certain portion of incident on the finger from the emitter of the sensor **120** may not be absorbed by the finger, but may instead be reflected off of the finger. In addition, a certain portion of attenuated light will exit the other side of the finger. Because the cartridge **1100** generally envelopes the finger and is absorptive of the emitted light, light which is not absorbed by the finger or otherwise may advantageously escape into the region between the cartridge and the finger will be substantially absorbed by the cartridge **1100** due to its absorptive properties, reducing the effect of light bounce.

Alternative configurations of the cartridge **1100** are possible as will be recognized by skilled artisans from the disclosure herein. For example, the cartridge **1100** body may comprise a substantially thin rubber material in some embodiments. The cartridge **1100** may include an adhesive proximate the perimeter of the opening **1102**, such as the self-adhesive strip **820** of the cartridge **800** of FIG. **8**, which may help create a seal between the finger and the cartridge **1100**. In other configurations, the perimeter of the opening may include an elastic band capable of creating an elastic seal between the finger and the cartridge **1100**. In certain embodiments, the cartridge **1100** or a portion thereof may

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include a different color such as a relatively light color, or may be translucent or partially translucent. In one configuration, the interior of the cartridge 1100 comprises a substantially dark color and the outer surface comprises another color, such as a light color. In yet another embodiment, the interior of the cartridge 110 comprises a substantially light color and the outer surface comprises a substantially dark color.

FIGS. 12A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure. The cartridge 1200 of FIG. 12 may be generally similar in structure and function to the cartridge 1100 of FIG. 11. For example, the cartridge 1200 includes an opening 1202, a cavity 1208, an upper aperture 1204 and a lower aperture 1206. In addition, the cartridge 1200 comprises a color absorptive of the emitted light and advantageously reduces the effect of light bounce.

The cartridge 1200 may not fit in a uniformly snug manner with the upper and lower sensor housings 122, 124. For example, as the upper and lower sensor housings 122, 124 exert force directly on the outer surface of the cartridge 1200 and thus indirectly on the tissue site within the cartridge 1200, the sensor cartridge 1200 may flex and partially deform in response to this force. This flexing may create gaps between portions of the inner surfaces of the upper and lower sensor housings 122, 124 and the outer surface of the cartridge 1200. The cartridge 1200 includes connecting portions 1210, 1212 which are configured to bridge potential gaps between the outer surface of the cartridge 1200 and the inner surfaces of the upper and lower sensor housings 122, 124. Particularly, the connecting portions 1210, 1212 are configured to bridge these gaps substantially in the region of the perimeter of the apertures 1204, 1206, respectively. The connecting portions 1210, 1212 thereby create a seal between the perimeter of the apertures 1204, 1206 and the inner surfaces of the upper and lower sensor housings 122, 124, respectively. Channel regions are thus created whereby light from the emitter can travel from the emitter to the aperture 1204 and whereby attenuated light can travel from the tissue site to the detector. Thus, the connecting portion 1210 prevents light directed towards the upper aperture 1204 from the emitter from escaping into the region between the inner surface of the upper sensor housing 122 and the outer surface of the cartridge 1200. Likewise, the connecting portion 1212 prevents attenuated light exiting the tissue site towards the detector of the sensor 120 from prevented from escaping into the region between the inner surface of the lower sensor housing 124 and the cartridge 1200. Because light escaping into these regions may contribute to light bounce, as described above, the connecting portions 1210, 1212 advantageously provide further reduction of light bounce. Moreover, the connecting portions 1210, 1212 generally cause a greater percentage of light from the emitter to be directly incident on the tissue site and a cause greater percentage of attenuated light exiting from the tissue site to be directly incident on the detector. The connecting portions 1210, 1212 thus can provide for increased measurement accuracy, improved calibration and efficiency of sensor operation, among other advantages.

The connecting portions 1210, 1212 of the illustrated embodiment include four panels each extending from one side of the perimeter of the upper and lower apertures 1204, 1206 to form raised rectangular borders around the apertures 1204, 1206. The connecting portions 1210, 1212 are configured to contact the interior surfaces of the upper and lower sensor housings 122, 124 around the emitter and the detector, respectively. The connecting portions 1210, 1212

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include a flexible material such as a rubber or plastic which conforms to the interior surfaces of the upper and lower sensor housings 122, 124, respectively. As such, the connecting portion 1210 helps to create a seal between the emitter of the sensor 120 and the cartridge as described above, thereby reducing light bounce and providing for increased measurement accuracy. Likewise, the connecting portion 1212 helps to create a seal between the detector of the sensor 120 and the cartridge 1200 as described above, thereby further reducing light bounce and improving measurement accuracy.

As will be appreciated by those of skill in the art from the disclosure provided herein, alternative configurations of the cartridge 1200 are possible. For example, in certain embodiments, one or more of the connecting portions 1210, 1212 may comprise a rigid material. The connecting portions 1210, 1212 may mate with corresponding features of the interior surfaces of the upper and lower sensor housings 122, 124, respectively. For example, the connecting portions 1210, 1212 may form raised features such as in the illustrated embodiment which fit into corresponding female portions, such as recesses in the interior surfaces of the sensor housings 122, 124. In various embodiments, snap-fit, friction-fit, and other mating mechanisms may be employed. Certain features may be reversed in some embodiments. For example, one or more of the interior surfaces of the sensor housings 122, 124 may include male portions and the connecting portions 1210, 1212 may include female portions. In certain embodiments, connecting portions such as the connecting portions 1210, 1212 may be used on other cartridges described herein, such as, for example, the cartridge 110 of FIGS. 1-2, or the cartridge 300 of FIG. 3.

Various sensor cartridges and covers have been disclosed in detail in connection with various embodiments. These embodiments are disclosed by way of examples only and are not to limit the scope of the claims that follow. One of ordinary skill in the art will appreciate the many variations, modifications and combinations. For example, in one embodiment, the cartridge 110 of FIG. 2 includes a securing feature such as the securing feature 311 of the cartridge 300 of FIG. 3. In some embodiments, the cartridges 110, 300 of FIGS. 2 and 3 are configured to substantially envelope the tissue site in a manner similar to the cartridges 1100, 1200 of FIGS. 11 and 12. In various embodiments, any of the cartridges described throughout the disclosure, such as the cartridges 300, 510, 610, 1100, 1200 described with respect to FIGS. 3, 5, 6, 11 and 12, may include a film covering the respective apertures. For example, the film may include a film such as the one described above with respect to the apertures 115, 117 of the cartridge 110 of FIGS. 1-2. In addition, in various embodiments the sensor cartridges and covers are used with a sensor that may measure any type of physiological parameter. In various embodiments, the sensor cartridges and covers may be for any type of medical device.

What is claimed is:

1. A disposable sterile barrier configured to interpose a material between a tissue site and surfaces of a sensor configured to grasp the tissue site, the sensor configured to transmit optical radiation into the tissue site and to generate a signal responsive to the optical radiation after attenuation by pulsatile blood flow within the tissue site, the disposable sterile barrier comprising:

a barrier comprising a collapsible tube the barrier being made of a substantially optically transparent material;

a first portion of the collapsible tube including an open end, wherein the first portion is sized to receive a reusable optical sensor in the collapsible tube, and

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wherein at least a portion of the first portion is configured to be interposed between a tissue site and a first surface of the reusable optical sensor proximate an emitter of the reusable optical sensor; and

a second portion of the collapsible tube including a closed end, wherein the second portion is sized to receive the reusable optical sensor in the collapsible tube, and wherein at least a portion of the second portion is configured to be interposed between the tissue site and a second surface of the reusable optical sensor proximate a detector of the reusable optical sensor.

2. The disposable sterile barrier of claim 1, wherein: the barrier comprising the collapsible tube is configured to be placed over the reusable optical sensor and over at least a portion of a sensor cable extending from the reusable optical sensor; and

the barrier further comprises a fastener proximate the open end of the collapsible tube, the fastener configured to close the open end of the collapsible tube around the sensor cable.

3. The disposable sterile barrier according to claim 2, wherein the collapsible tube is capable of accommodating a first sensor housing comprising one of the first and second surfaces and to accommodate a second sensor housing comprising the other of the first and second surfaces so as to enclose the reusable optical sensor within the collapsible tube.

4. The disposable sterile barrier of claim 1, wherein the tissue site is at least a portion of a finger.

5. An optical sensing method of non-invasively measuring constituents of pulsatile blood flow within a tissue site without substantial risk of nosocomial infection by direct contact between a reusable optical sensor and the tissue site, the optical sensing method comprising:

providing a reusable optical sensor having an emitter disposed within a first housing and a detector disposed within a second housing, the emitter and detector in communication with a sensor cable, the first and second housings configured to be urged against opposite sides of a tissue site upon application of the reusable optical sensor to the tissue site, the emitter configured to transmit optical radiation having one or more predetermined wavelengths into the tissue site and the detector configured to receive the optical radiation after attenuation by the tissue site, the reusable optical sensor configured to generate one or more signals indicative of the attenuated radiation, the one or more signals transmitted via the sensor cable to a monitor

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configured to process the one or more signals to determine one or more physiological parameters of a patient; and

applying a barrier so as to make physiological measurements without direct contact between the tissue site and the reusable optical sensor, the barrier configured to be interposed between the tissue site and the first housing and configured to be interposed between the tissue site and the second housing upon application of the tissue site to the reusable optical sensor, the barrier comprising material substantially transparent so as not to substantially distort the physiological measurements, the barrier further comprising a collapsible tube comprising a first portion and a second portion, the first portion comprising an open end and sized to receive the reusable optical sensor, the second portion comprising a closed end and sized to receive the reusable optical sensor.

6. The optical sensing method of claim 5, further comprising disposing of the barrier after determining one or more physiological parameters of the patient.

7. The optical sensing method of claim 5, wherein applying further comprises:

providing the barrier comprising the collapsible tube; expanding the collapsible tube from a collapsed state to an un-collapsed state;

fitting the open end of the collapsible tube around one of the first sensor housing and the second sensor housing; moving the open end firstly along one of the first and second housings and then along the other of the first and second housings such that the reusable optical sensor and at least a portion of the sensor cable are enclosed within the tube; and

sealing the open end around the sensor cable.

8. The optical sensing method of claim 5, wherein the tissue site is at least a portion of a finger.

9. A disposable sterile barrier comprising:

an elongate tube comprising a cavity, an open end and a closed end, the open end comprising an opening large enough to accommodate a reusable optical sensor, the cavity sized to fully enclose the reusable optical sensor and at least a portion of a sensor cable extending from the optical sensor, the elongate tube comprising a material which is substantially optically transparent; and

a fastener disposed proximate the open end of the elongate tube and configured to seal the open end of the elongate tube around the sensor cable.

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